Databricks ML Associate

100 Exam-Style Questions

Practice Exam Deck

How to Use This Deck

100 questions, each on its own slide with four options (A–D).

The **next slide** gives the correct answer and explains:

- why the correct option is right;
- why each wrong option is wrong.

Domains: Databricks ML (Q1–38), ML Workflows (Q39–57), Model Development (Q58–88), Model Deployment (Q89–100).

Question 1

A machine learning engineer created a Feature Table `new_table` using Feature Store Client `fs` and specified a metadata description. Which line of code returns that description?

A. `fs.create_training_set("new_table")`

B. `fs.get_table("new_table").description`

C. `fs.get_table("new_table").load_df()`

D. `fs.get_table("new_table")`

Answer 1

Correct: B — `get_table()` returns the table object and `.description` reads its metadata.

Wrong:

A creates a training set, not metadata retrieval.

C loads data into a DataFrame.

D returns the object but not the description.

Question 2

What is the primary advantage of creating Feature Store tables at the account level in Unity Catalog rather than at the workspace level?

A. Faster model training

B. Cross-workspace governance, discoverability, and access control

C. Automatic feature engineering

D. Lower storage cost

Answer 2

Correct: B — account-level tables are governed and discoverable across workspaces.

Wrong:

A is not a feature of account-level tables.

C is false; feature engineering is manual.

D is not the primary advantage.

Question 3

Which client is used to create and manage Feature Store tables in Unity Catalog?

A. `FeatureStoreClient`
B. `FeatureEngineeringClient`
C. `UnityCatalogClient`
D. `MLflowClient`

Answer 3

Correct: B — `FeatureEngineeringClient` is the Unity Catalog Feature Store client.

Wrong:

A is the legacy workspace client.

C is not a real client.

D manages MLflow, not feature tables.

Question 4

Which method creates a feature table with `FeatureEngineeringClient`?

A. `fe.create_table()`
B. `fe.write_table()`
C. `fe.create_training_set()`
D. `fe.log_model()`

Answer 4

Correct: A — `create_table()` creates the table.

Wrong:

B writes features to an existing table.

C builds a training set.

D logs a model.

Question 5

Which method writes features to an existing feature table?

A. `fe.create_table()`

B. `fe.write_table()`

C. `fe.score_batch()`

D. `fe.get_table()`

Answer 5

Correct: B — `write_table()` writes features.

Wrong:

A creates a table.

C scores a batch.

D retrieves a table.

Question 6

What does `fe.create_training_set()` provide that a plain DataFrame join does not?

A. Automatic one-hot encoding

B. Point-in-time lookups aligned to the label timestamp

C. Model registration

D. Hyperparameter tuning

Answer 6

Correct: B — training sets use point-in-time lookups aligned to label timestamps.

Wrong:

A is a separate preprocessing step.

C is done by `register_model`.

D is done by Hyperopt.

Question 7

A model performs well offline but degrades in production. Review shows a feature was computed using data available only after the label outcome. What is the diagnosis?

A. Underfitting

B. Training-serving skew / target leakage

C. Class imbalance

D. Concept drift

Answer 7

Correct: B — post-label data causes leakage and skew.

Wrong:

A would show poor training performance.

C does not explain post-label features.

D is a change over time, not a data leak.

Question 8

What is the difference between online and offline feature tables?

A. Online is for training; offline is for serving

B. Online serves low-latency real-time inference; offline is for batch training and scoring

C. They are identical

D. Offline is deprecated

Answer 8

Correct: B — online tables serve low-latency inference; offline tables are for batch.

Wrong:

A reverses the roles.

C is false.

D is false; offline tables are not deprecated.

Question 9

A feature-aware model was logged with `fs.log_model()` and a `batch_df` contains only the primary key `customer_id`. Which call generates predictions by automatically joining the required features?

A. `model.predict(batch_df)`

B. `fs.score_batch(model_uri, batch_df)`

C. `mlflow.pyfunc.load_model(model_uri).predict(batch_df)`

D. `batch_df.join(feature_table)`

Answer 9

Correct: B — `fs.score_batch()` uses logged feature metadata to join features automatically.

Wrong:

A assumes the features are already present.

C also assumes features are in the DataFrame.

D is a plain join without automatic lookup.

Question 10

Which namespace format is correct for a feature table in Unity Catalog?

A. `catalog.schema.table`

B. `database.table`

C. `workspace.table`

D. `hive_metastore.table`

Answer 10

Correct: A — Unity Catalog uses `catalog.schema.table`.

Wrong:

B is a two-level format.

C is not a Unity Catalog format.

D is a legacy Hive reference.

Question 11

Which MLflow API call manually logs a metric?

A. `mlflow.log_metric()`

B. `mlflow.log_param()`

C. `mlflow.log_artifact()`

D. `mlflow.set_tag()`

Answer 11

Correct: A — `log_metric()` logs metrics.

Wrong:

B logs parameters.

C logs artifacts.

D sets tags.

Question 12

Which MLflow API call registers a model in Unity Catalog?

A. `mlflow.log_model()`

B. `mlflow.register_model()`

C. `mlflow.save_model()`

D. `mlflow.start_run()`

Answer 12

Correct: B — `register_model()` registers a model.

Wrong:

A logs a model artifact.

C saves locally.

D starts a run.

Question 13

What replaced the old Staging and Production stages for model promotion in Unity Catalog?

A. Tags only

B. Aliases such as `@champion`

C. Notebook revisions

D. Git branches

Answer 13

Correct: B — aliases replace stages.

Wrong:

A tags do not replace promotion.

C tracks code, not model stages.

D tracks code versions.

Question 14

How do you promote a challenger model to champion using aliases?

A. Delete the old model version
B. Move the `champion` alias to the new model version
C. Rename the model
D. Create a new experiment

Answer 14

Correct: B — promotion is moving the alias.

Wrong:

A deletes history.

C does not change deployment.

D is unrelated.

Question 15

What does `mlflow.search_runs()` return?

A. A single run ID

B. A pandas DataFrame of runs that can be sorted and filtered

C. A trained model

D. A list of experiments

Answer 15

Correct: B — it returns a pandas DataFrame of runs.

Wrong:

A is too narrow.

C is a model object.

D lists experiments, not runs.

Question 16

A run ID is stored in `run_id` and the logged model name is `"model"`. Which line registers it as `"best_model"`?

A. `mlflow.register_model(run_id, "best_model")`

B. `mlflow.register_model(f"runs:/{run_id}/model", "best_model")`

C. `mlflow.register_model(f"runs:/{run_id}/model")`

D. `mlflow.register_model(f"runs:/{run_id}/best_model", "model")`

Answer 16

Correct: B — the URI must include `runs:/` and the artifact path.

Wrong:

A passes only the run ID, not a model URI.

C omits the registry name.

D swaps artifact path and name.

Question 17

A data scientist wants one parent run for tuning and a child run per trial, starting runs manually. What should they do?

A. Turn on Databricks Autologging

B. Specify `nested=True` when starting each child run

C. Start each child run inside the parent's code block without `nested`

D. Give child runs the same experiment ID

Answer 17

Correct: B — child runs must be started with `nested=True`.

Wrong:

A does not create nesting.

C raises an error.

D only groups runs in the same experiment.

Question 18

Which UI element on the MLflow experiment page opens the notebook that created a run?

A. The `MLmodel` artifact
B. The `Models` link
C. The `Source` link
D. The `Start Time` link

Answer 18

Correct: C — the Source link opens the notebook snapshot.

Wrong:

A is model metadata.

B links to the registry.

D opens run details.

Question 19

Which function searches runs ordered by a metric ascending to find the lowest RMSE?

A. `mlflow.search_runs(experiment_id, order_by=["metrics.rmse DESC"])`

B. `mlflow.best_run(experiment_id, order_by=["metrics.rmse"])`

C. `mlflow.search_runs(experiment_id, order_by=["metrics.rmse ASC"])`

D. `mlflow.search_runs(experiment_id, filter="rmse < 1")`

Answer 19

Correct: C — ascending order puts the lowest RMSE first.

Wrong:

A orders descending.

B is not the API.

D filters rather than sorts.

Question 20

On which page do you transition a registered model version from Staging to Production?

A. The MLflow Model Registry home page

B. The experiment page

C. The model version page

D. The model page only

Answer 20

Correct: C — stage transitions happen on the model version page.

Wrong:

A lists all models.

B shows runs.

D lists versions but not the direct controls.

Question 21

Which MLflow API call logs an artifact?

A. `mlflow.log_metric()`

B. `mlflow.log_param()`

C. `mlflow.log_artifact()`

D. `mlflow.set_tag()`

Answer 21

Correct: C — `log_artifact()` logs files.

Wrong:

A logs metrics.

B logs parameters.

D sets tags.

Question 22

What is the simplest Databricks-native way to make every training run reproducible months later?

A. Save notebooks to a local drive

B. Use MLflow tracking to log parameters, metrics, artifacts, and the model

C. Print metrics to stdout

D. Rely on cluster logs

Answer 22

Correct: B — MLflow tracking logs all run data for reproducibility.

Wrong:

A is not centralized.

C is not durable.

D may be ephemeral.

Question 23

What does Databricks AutoML produce besides a trained model?

A. Only a model artifact

B. A leaderboard, editable notebooks, and an automated EDA notebook

C. A production endpoint

D. A data warehouse

Answer 23

Correct: B — AutoML returns a leaderboard, notebooks, and EDA.

Wrong:

A is incomplete.

C is a deployment step.

D is not produced.

Question 24

Which API runs classification AutoML?

A. `databricks.automl.classify()`

B. `databricks.automl.regress()`

C. `databricks.automl.forecast()`

D. `databricks.automl.train()`

Answer 24

Correct: A — `classify()` runs classification AutoML.

Wrong:

B is regression.

C is forecasting.

D is not an API.

Question 25

AutoML is least appropriate when:

A. You need a quick baseline

B. You need a highly customized deep learning architecture

C. You want automatic feature engineering

D. You want to compare multiple algorithms

Answer 25

Correct: B — AutoML is less suited to highly custom deep learning.

Wrong:

A is a strength of AutoML.

C is a strength of AutoML.

D is a strength of AutoML.

Question 26

Which step must a data scientist perform outside an AutoML experiment?

A. Model tuning

B. Model evaluation

C. Model deployment

D. Exploratory data analysis

Answer 26

Correct: C — deployment is outside AutoML.

Wrong:

A is automated.

B is automated.

D is automated via the EDA notebook.

Question 27

How do you begin using Databricks Runtime ML?

A. Add an init script line

B. Check a Databricks Runtime ML box

C. Select a Runtime ML version from the Databricks Runtime Version dropdown

D. Set `runtime-version="ml"` in the Spark session

Answer 27

Correct: C — Runtime ML is selected from the runtime dropdown.

Wrong:

A does not select the runtime.

B there is no such box.

D does not change the runtime.

Question 28

Why is Databricks Runtime ML convenient for MLflow users?

A. It removes the need for a cluster
B. It preinstalls MLflow and other ML libraries
C. It disables logging
D. It only supports R

Answer 28

Correct: B — Runtime ML preinstalls MLflow and ML libraries.

Wrong:

A is false.

C is false.

D is false; it supports Python and more.

Question 29

Which evaluation metric is not suitable for AutoML regression experiments?

A. F1
B. R-squared
C. MAE
D. MSE

Answer 29

Correct: A — F1 is for classification, not regression.

Wrong:

B is a regression metric.

C is a regression metric.

D is a regression metric.

Question 30

Which file typically defines a model's format and dependencies in an MLflow run?

A. `requirements.txt`

B. `MLmodel`

C. `README.md`

D. `conda.yaml` only

Answer 30

Correct: B — `MLmodel` defines format and dependencies.

Wrong:

A is a dependency list.

C is documentation.

D is one dependency spec, not the format file.

Question 31

What does `fs.log_model()` add to a model artifact?

A. Hyperparameters

B. Feature lookup metadata for automatic joins at inference

C. A confusion matrix

D. A serving endpoint

Answer 31

Correct: B — it packages feature lookup metadata.

Wrong:

A is logged separately.

C is not added by this call.

D is created later.

Question 32

Which is a benefit of registering models in Unity Catalog over the workspace registry?

A. Faster training

B. Lineage, governance, and cross-workspace access

C. Automatic hyperparameter tuning

D. Free storage

Answer 32

Correct: B — Unity Catalog adds lineage, governance, and access.

Wrong:

A is false.

C is false.

D is false.

Question 33

What does an MLflow experiment group together?

A. Runs of a related ML task
B. Clusters
C. Notebooks only
D. Feature tables

Answer 33

Correct: A — experiments group related runs.

Wrong:

B clusters are compute.

C notebooks are source code.

D feature tables are data assets.

Question 34

Which API call starts an MLflow run?

A. `mlflow.start_run()`

B. `mlflow.log_run()`

C. `mlflow.open_run()`

D. `mlflow.new_run()`

Answer 34

Correct: A — `start_run()` starts a run.

Wrong:

B is not an API.

C is not an API.

D is not an API.

Question 35

Which is true about aliases in the Unity Catalog model registry?

A. An alias can point to many versions at once

B. An alias points to one model version and can be moved

C. Aliases are immutable

D. Aliases replace experiments

Answer 35

Correct: B — an alias points to one version and can be moved.

Wrong:

A is false.

C aliases are movable.

D aliases are model-level, not experiment-level.

Question 36

Which object stores the actual model files and metadata for a run?

A. The artifact store

B. The parameter log

C. The metric log

D. The experiment tag

Answer 36

Correct: A — artifacts live in the artifact store.

Wrong:

B stores parameters.

C stores metrics.

D stores tags.

Question 37

Why should secrets not be logged as MLflow parameters?

A. Parameters are stored in plain text and visible to experiment users
B. Parameters are not persisted
C. Parameters are encrypted by default
D. Parameters are deleted after the run

Answer 37

Correct: A — parameters are stored in plain text.

Wrong:

B is false; they are persisted.

C is false.

D is false.

Question 38

Which is a primary benefit of using MLflow Model Registry?

A. It replaces all experiments

B. It manages model versions, aliases, and lifecycle

C. It trains models

D. It stores feature tables

Answer 38

Correct: B — the registry manages versions, aliases, and lifecycle.

Wrong:

A experiments still exist.

C it does not train models.

D feature tables are in Feature Store.

Question 39

A Spark DataFrame `spark_df` must keep only rows where `price > 0`. Which code block works?

A. `spark_df[spark_df["price"] > 0]`

B. `spark_df.filter(col("price") > 0)`

C. `SELECT * FROM spark_df WHERE price > 0`

D. `spark_df.loc[spark_df["price"] > 0,:]`

Answer 39

Correct: B — `filter(col(...))` is the PySpark way.

Wrong:

A is pandas-like and not PySpark-idiomatic.

C is SQL, not DataFrame code.

D is pandas syntax.

Question 40

Which line returns count, mean, standard deviation, min, max, and quantiles for numeric columns?

A. `spark_df.summary()`

B. `spark_df.stats()`

C. `spark_df.describe().head()`

D. `spark_df.printSchema()`

Answer 40

Correct: A — `summary()` returns count, mean, std, min, max, and quantiles.

Wrong:

B is not an API.

C lacks quantiles.

D prints the schema.

Question 41

How do you remove outliers using IQR in Spark?

A. Drop the top 10% of rows

B. Filter values outside $[Q1 - 1.5 \cdot IQR,\ Q3 + 1.5 \cdot IQR]$

C. Filter values outside $[Q1 - IQR,\ Q3 + IQR]$

D. Replace all values with the mean

Answer 41

Correct: B — the $1.5 \times IQR$ rule is standard.

Wrong:

A is not IQR-based.

C uses a wrong threshold.

D does not remove outliers.

Question 42

When is median imputation preferred over mean imputation?

A. When data is symmetric

B. When data is skewed or contains extreme outliers

C. When there are no missing values

D. Always

Answer 42

Correct: B — median is robust to skew and outliers.

Wrong:

A mean is fine for symmetric data.

C no imputation is needed.

D is false.

Question 43

A data scientist replaces missing values with the median. A colleague says valuable information is lost. Which approach preserves the most information?

A. Impute with the mean instead

B. Let the algorithm handle missing values

C. Drop all features with missing values

D. Add a binary indicator feature marking which values were imputed

Answer 43

Correct: D — indicator features preserve missingness information.

Wrong:

A still loses missingness information.

B not all algorithms handle missing values well.

C discards features entirely.

Question 44

One-hot encoding is least appropriate for:

A. Nominal categorical features
B. Very high-cardinality categorical features
C. Low-cardinality categorical features
D. Tree-based models

Answer 44

Correct: B — high cardinality causes dimensionality explosion.

Wrong:

A is a typical use case.

C is a typical use case.

D trees can use one-hot but are not the issue.

Question 45

A feature repository one-hot encodes all categorical variables. Why is this problematic?

A. One-hot encoding is not supported by most libraries

B. It depends on target values

C. It is computationally intensive and should be done on small samples

D. It is a potentially problematic strategy for some ML algorithms due to dimensionality and sparsity

Answer 45

Correct: D — premature one-hot can hurt some algorithms.

Wrong:

A is false.

B one-hot does not use target values.

C is not the main issue.

Question 46

How do you one-hot encode in Spark ML?

A. `OneHotEncoder` alone
B. `StringIndexer` then `OneHotEncoder`
C. `VectorAssembler` then `StringIndexer`
D. `Imputer` then `OneHotEncoder`

Answer 46

Correct: B — Spark ML uses StringIndexer then OneHotEncoder.

Wrong:

A needs indexed input first.

C is wrong order.

D imputation is unrelated.

Question 47

What does `VectorAssembler` do?

A. Combines multiple columns into a single feature vector

B. Splits a vector into columns

C. Encodes strings

D. Imputes missing values

Answer 47

Correct: A — VectorAssembler combines columns into a vector.

Wrong:

B is the opposite operation.

C is StringIndexer.

D is Imputer.

Question 48

A data scientist creates an `Imputer` and calls `transform` directly on it. Why does it fail?

A. It does not impute both training and test sets

B. `inputCols` and `outputCols` must be identical

C. It should call `fit` instead of `transform`

D. It does not fit the imputer to create an `ImputerModel`

Answer 48

Correct: D — the imputer must be fit first.

Wrong:

A is not the immediate cause.

B is false; columns can differ.

C is imprecise; you fit then transform.

Question 49

When is log transformation appropriate?

A. For right-skewed positive features
B. For left-skewed negative features
C. For categorical features
D. For binary features

Answer 49

Correct: A — log transform suits right-skewed positive data.

Wrong:

B is not a typical use case.

C is for categorical data.

D is for binary data.

Question 50

A pandas notebook must scale to big data with the least refactoring effort. What approach is recommended?

A. Refactor to process data in parallel
B. Refactor to the PySpark DataFrame API
C. Refactor to the Scala Dataset API
D. Refactor to Spark SQL
E. Refactor to the pandas API on Spark

Answer 50

Correct: E — pandas API on Spark minimizes code changes.

Wrong:

A–D require larger rewrites.

Question 51

Which function applies a Python function to each group of a grouped DataFrame and returns a DataFrame?

A. `applyInPandas`

B. `groupedApplyInPandas`

C. `mapInPandas`

D. `predict`

Answer 51

Correct: A — `applyInPandas` applies per group and returns a DataFrame.

Wrong:

B is not an API.

C maps partitions, not groups.

D is not an API.

Question 52

A data scientist wants group-specific model inference in parallel. Which call fills the blank in `df.groupby("device_id").___(apply_model, schema=...)` ?

A. `applyInPandas`

B. `groupedApplyInPandas`

C. `mapInPandas`

D. `predict`

Answer 52

Correct: A — `applyInPandas` is the grouped API.

Wrong:

B is not an API.

C maps partitions.

D is not an API.

Question 53

Which imputation strategy is most appropriate for a numeric feature with extreme outliers?

A. Mean

B. Median

C. Mode

D. Zero

Answer 53

Correct: B — median is robust to outliers.

Wrong:

A is distorted by outliers.

C is for categorical data.

D can bias the distribution.

Question 54

What does a boxplot show?

A. Only the mean
B. Median, quartiles, and outliers
C. Correlation
D. Class balance

Answer 54

Correct: B — boxplots show median, quartiles, and outliers.

Wrong:

A is incomplete.

C is a scatter/heatmap feature.

D is a bar chart feature.

Question 55

Which technique helps detect multicollinearity?

A. Variance Inflation Factor
B. Silhouette score
C. ROC AUC
D. Confusion matrix

Answer 55

Correct: A — VIF detects multicollinearity.

Wrong:

B measures clustering quality.

C measures classification ranking.

D evaluates classification outcomes.

Question 56

Why is stratification important for classification splits?

A. It preserves class proportions in each split
B. It increases training speed
C. It removes outliers
D. It ensures reproducibility only

Answer 56

Correct: A — stratification preserves class proportions.

Wrong:

B is not its purpose.

C is a different preprocessing step.

D is not its main purpose.

Question 57

Which is a symptom of an imbalanced dataset?

A. A 50/50 class split

B. A 99% negative / 1% positive split

C. A continuous target with normal distribution

D. Missing values

Answer 57

Correct: B — 99/1 is severe imbalance.

Wrong:

A is balanced.

C is a regression target property.

D is missingness, not imbalance.

Question 58

Which algorithm typically uses bagging?

A. Gradient boosted trees

B. K-means

C. Random forest

D. Linear regression

Answer 58

Correct: C — Random forest uses bagging.

Wrong:

A uses boosting.

B is clustering.

D is a single model.

Question 59

Spark ML linear regression on large data uses which approach for distributed training?

A. Logistic regression
B. Spark ML cannot distribute linear regression
C. Iterative optimization such as L-BFGS
D. Least-squares normal equations

Answer 59

Correct: C — large data uses iterative optimizers like L-BFGS.

Wrong:

A is a classification algorithm.

B is false.

D is limited to ≤ 4096 features.

Question 60

Why can a Spark ML decision tree differ from a scikit-learn tree with identical data and hyperparameters?

A. Spark ML tests every feature
B. Spark ML automatically prunes overfit trees
C. Spark ML tests more split candidates
D. Spark ML tests a random sample of features
E. Spark ML tests binned feature values as representative split candidates

Answer 60

Correct: E — Spark bins continuous features for distributed efficiency.

Wrong:

A is false.

B pruning is user-controlled.

C it tests fewer, binned candidates.

D random feature sampling is for Random Forest.

Question 61

What does Hyperopt's `fmin` do?

A. Minimizes an objective function over a search space

B. Maximizes accuracy only

C. Trains a model

D. Logs a metric

Answer 61

Correct: A — `fmin` minimizes the objective.

Wrong:

B it is not limited to accuracy.

C it calls a user objective.

D it does not log metrics.

Question 62

What does `SparkTrials` do?

A. Distributes Hyperopt trials across a Spark cluster
B. Runs trials on the driver only
C. Saves models to the registry
D. Encodes categorical features

Answer 62

Correct: A — `SparkTrials` distributes trials.

Wrong:

B is the opposite.

C is model registration.

D is a preprocessing step.

Question 63

What is the role of `tpe.suggest` in Hyperopt?

A. Evaluates the objective function

B. Specifies the MLflow experiment

C. Suggests new hyperparameter configurations based on previous results

D. Parallelizes the tuning process

Answer 63

Correct: C — `tpe.suggest` proposes new configurations.

Wrong:

A is the objective function's job.

B is MLflow integration.

D is `SparkTrials`.

Question 64

A team has 8 workers and 5 hyperparameters. Which approach best leverages the cluster?

A. Grid search on the driver

B. Sequential manual tuning

C. Hyperopt with `SparkTrials` and TPE

D. Random search on the driver

Answer 64

Correct: C — Hyperopt with `SparkTrials` and TPE leverages the cluster.

Wrong:

A–B–D are inefficient or single-node.

Question 65

With 3-fold CV and a grid of 3×2 hyperparameter values, how many models are trained in total?

A. 6

B. 9

C. 18

D. 36

Answer 65

Correct: C — $3 \times 2 \times 3 = 18$.

Wrong:

A is just the combinations.

B is wrong arithmetic.

D doubles the folds incorrectly.

Question 66

With 3-fold CV and a grid of 3×2 values, how many models can be trained in parallel?

A. 3

B. 5

C. 6

D. 18

Answer 66

Correct: C — six combinations can run in parallel.

Wrong:

A is the number of folds.

B is arbitrary.

D is the total, not parallel.

Question 67

What is a primary advantage of Hyperopt over manual tuning or grid search?

A. It efficiently explores the search space using algorithms like TPE
B. Manual tuning is faster
C. Grid search is more reliable
D. Hyperopt needs more evaluations

Answer 67

Correct: A — Hyperopt explores more efficiently with TPE.

Wrong:

B is false.

C is false.

D is false.

Question 68

When is `StringIndexer` used in a pipeline?

A. To convert textual or categorical values to numeric indices
B. To impute missing values
C. To scale numeric features
D. To assemble vectors

Answer 68

Correct: A — StringIndexer converts categories to indices.

Wrong:

B is Imputer.

C is StandardScaler.

D is VectorAssembler.

Question 69

Why might a train-validation split be more efficient than k-fold CV?

A. It requires testing fewer models, saving compute
B. It reduces hyperparameter values to test
C. It guarantees reproducibility
D. It ensures bias-free training

Answer 69

Correct: A — a split trains fewer models than k-fold CV.

Wrong:

B is false.

C is not guaranteed.

D is false.

Question 70

When using a pipeline and a cross-validator, where should they be placed?

A. Always put the cross-validator inside the pipeline
B. If the pipeline includes estimators, put the entire pipeline inside the cross-validator
C. If concerned about leakage, put the pipeline inside the cross-validator
D. Always put the pipeline inside the cross-validator

Answer 70

Correct: B — put the pipeline inside the CV when it contains estimators.

Wrong:

A causes leakage.

C is only the leakage special case.

D is too absolute; estimators dictate placement.

Question 71

A health organization wants to maximize the number of positive infection cases identified. Which metric should be used?

A. RMSE

B. Precision

C. Area under the residual operating curve

D. Accuracy

E. Recall

Answer 71

Correct: E — recall maximizes identification of positives.

Wrong:

A is regression.

B focuses on false positives.

C is not a real metric.

D is misleading under imbalance.

Question 72

When should F1 be used over accuracy?

A. When classes are balanced

B. When there are more than two classes

C. When there is significant imbalance and avoiding false negatives is a priority

D. When true positives and true negatives are equally important

Answer 72

Correct: C — F1 is for imbalance with false-negative priority.

Wrong:

A accuracy is fine when balanced.

B F1 extends to multiclass but is not uniquely for it.

D accuracy captures both TPs and TNs.

Question 73

A fraud dataset has a 1% positive class and a model reports 99% accuracy. What should you do?

A. Deploy it

B. Evaluate precision, recall, F1, and PR-AUC

C. Increase epochs

D. Add features

Answer 73

Correct: B — evaluate imbalance-aware metrics.

Wrong:

A accuracy is misleading.

C does not fix the metric.

D is premature.

Question 74

What does $R^2 = 0.85$ mean?

A. 85% of predictions are correct

B. The model explains 85% of the variance in the target

C. The model has 85% precision

D. The RMSE is 0.85

Answer 74

Correct: B — R^2 is the fraction of variance explained.

Wrong:

A is classification accuracy.

C is precision.

D is RMSE, a different metric.

Question 75

Why might RMSE be higher than MAE?

A. RMSE penalizes large errors more heavily
B. MAE squares errors
C. RMSE ignores outliers
D. MAE is in different units

Answer 75

Correct: A — RMSE penalizes large errors more.

Wrong:

B MAE uses absolute values.

C RMSE is more outlier-sensitive.

D both are in target units.

Question 76

Training accuracy is 99.5% and validation accuracy is 72%. What is the diagnosis and fix?

A. Underfitting; increase depth

B. Overfitting; reduce complexity or add regularization

C. Good model; deploy

D. Need more test data

Answer 76

Correct: B — a large train-validation gap means overfitting.

Wrong:

A underfitting shows poor train accuracy.

C the gap shows poor generalization.

D more test data does not fix overfitting.

Question 77

What is the effect of increasing model complexity on bias and variance?

A. Bias decreases, variance increases

B. Bias increases, variance decreases

C. Both decrease

D. Both increase

Answer 77

Correct: A — complexity lowers bias and raises variance.

Wrong:

B is the reverse.

C is false.

D is false.

Question 78

A log(price) model is evaluated against actual price values and its RMSE is much larger than a price model. Which explanation is invalid?

A. The second model is much more accurate

B. They failed to exponentiate predictions before computing RMSE

C. They failed to take the log of predictions in the first model

D. The first model is much more accurate

E. RMSE is invalid for regression

Answer 78

Correct: E — RMSE is valid for regression.

Wrong:

A–D are possible explanations.

Question 79

What is the 3-fold CV RMSE given fold RMSEs 10.0, 12.0, and 17.0?

A. 13.0

B. 17.0

C. 12.0

D. 39.0

Answer 79

Correct: A — $(10 + 12 + 17)/3 = 13.0$.

Wrong:

B is the max fold.

C is one fold.

D is the sum.

Question 80

What happens if the model is placed as a stage inside a pipeline and the cross-validator is the final stage?

A. Training takes longer

B. Feature engineering is computed using validation data, causing leakage

C. CV is no longer reproducible

D. The model is refit once more per fold

Answer 80

Correct: B — feature engineering on validation data leaks.

Wrong:

A is not the main consequence.

C CV remains reproducible.

D is not the issue.

Question 81

What does `CrossValidator` do in Spark ML?

A. Performs k-fold CV with a `ParamGridBuilder`
B. Trains one model
C. Clusters data
D. Imputes missing values

Answer 81

Correct: A — `CrossValidator` does k-fold CV with a parameter grid.

Wrong:

B is a single estimator.

C is clustering.

D is Imputer.

Question 82

When is a train-validation split preferable to k-fold CV?

A. When high interpretability is required
B. When compute is limited, especially for large data or complex models
C. When the algorithm is highly parallelizable
D. When compute is unlimited

Answer 82

Correct: B — a split is cheaper than k-fold CV.

Wrong:

A is unrelated.

C favors CV.

D favors CV.

Question 83

Which technique mitigates overfitting?

A. Feature engineering and dimensionality reduction
B. Increasing model complexity
C. Using only one technique such as data augmentation
D. All of the above

Answer 83

Correct: A — feature engineering and dimensionality reduction mitigate overfitting.

Wrong:

B worsens overfitting.

C is insufficient alone.

D is false.

Question 84

What is k-Nearest Neighbors' key hyperparameter tradeoff?

A. Small k gives low bias and high variance; large k gives high bias and low variance

B. Small k gives high bias; large k gives high variance

C. k has no effect

D. k only affects training time

Answer 84

Correct: A — small k means low bias and high variance.

Wrong:

B is reversed.

C is false.

D is false.

Question 85

Which kernel is commonly used for non-linear SVM classification?

A. Linear

B. RBF

C. Identity

D. Zero

Answer 85

Correct: B — RBF is common for non-linear SVM.

Wrong:

A is linear.

C is not a kernel.

D is not a kernel.

Question 86

What is gradient boosting?

A. Trees trained in parallel on bootstrap samples

B. Trees trained sequentially, each correcting previous errors

C. A clustering method

D. A dimensionality reduction method

Answer 86

Correct: B — boosting trains sequentially.

Wrong:

A is bagging.

C is clustering.

D is PCA.

Question 87

What is the role of `max_depth` in tree models?

A. Limits tree depth to control complexity and overfitting

B. Sets the number of trees

C. Sets the learning rate

D. Sets the number of features

Answer 87

Correct: A — `max_depth` limits depth.

Wrong:

B is `num_trees`.

C is `learning_rate`.

D is `max_features`.

Question 88

Which metric is most appropriate for a regression model with outliers?

A. RMSE

B. MAE

C. Accuracy

D. F1

Answer 88

Correct: B — MAE is less sensitive to outliers than RMSE.

Wrong:

A RMSE is more outlier-sensitive.

C is classification.

D is classification.

Question 89

What are the three model serving approaches tested on the exam?

A. Batch, real-time, and streaming

B. Batch, offline, and online

C. Training, validation, and testing

D. SQL, Python, and Scala

Answer 89

Correct: A — batch, real-time, and streaming.

Wrong:

B uses non-exam terms.

C is the ML lifecycle.

D is programming languages.

Question 90

A scikit-learn model must score 100 million Delta records efficiently. What is the best approach?

A. Load into pandas on the driver and loop

B. Use `mlflow.pyfunc.spark_udf()` to distribute inference

C. Export to CSV and score externally

D. Create a real-time endpoint and call per record

Answer 90

Correct: B — `spark_udf` distributes inference.

Wrong:

A does not scale.

C is manual.

D is high-latency per record.

Question 91

How is streaming inference performed in Databricks?

A. Using Delta Live Tables

B. Using a SQL Warehouse

C. Using the driver only

D. Using a pandas UDF on a single node

Answer 91

Correct: A — streaming inference uses Delta Live Tables.

Wrong:

B is SQL only.

C does not scale.

D is not streaming inference.

Question 92

How do you split traffic between real-time serving endpoints?

A. Configure traffic splitting on the serving endpoint
B. Duplicate the model
C. Use a SQL query
D. Rename the endpoint

Answer 92

Correct: A — traffic splitting is configured on the endpoint.

Wrong:

B does not split traffic.

C is not routing.

D does not split traffic.

Question 93

What is the primary purpose of a real-time serving endpoint?

A. Batch scoring

B. Low-latency predictions accessible via REST API

C. Training models

D. Data storage

Answer 93

Correct: B — real-time endpoints serve low-latency REST predictions.

Wrong:

A is batch inference.

C is training.

D is storage.

Question 94

A team replaces one model with three new models of nearly equal latency. When is the new solution less efficient during inference?

A. When if-else logic picks a model per prediction

B. When average model latency exceeds the original model's size

C. When fewer features are used

D. When every record must be scored by all three models

Answer 94

Correct: D — scoring every record with all three multiplies cost.

Wrong:

A only one model runs per record.

B is about model size, not inference count.

C fewer features reduce cost.

Question 95

What is a batch inference job in Databricks?

A. Scoring a large dataset offline and writing predictions to a table

B. A real-time REST endpoint

C. A training job

D. A cluster configuration

Answer 95

Correct: A — batch inference scores offline and writes predictions.

Wrong:

B is real-time serving.

C is training.

D is compute setup.

Question 96

Which compute type is best for Spark SQL data prep plus Spark ML training in a notebook?

A. Single Node cluster
B. Standard (all-purpose) cluster
C. SQL Warehouse
D. None of these

Answer 96

Correct: B — a Standard cluster supports SQL and Spark ML.

Wrong:

A is single-node.

C is SQL-only.

D is false.

Question 97

Why is model monitoring important in production?

A. Data distributions and relationships can drift over time

B. Once deployed, models never degrade

C. It replaces evaluation

D. It improves training speed

Answer 97

Correct: A — drift degrades deployed models.

Wrong:

B is false.

C evaluation is separate.

D monitoring is runtime, not training.

Question 98

What is data drift versus concept drift?

A. Data drift changes input distributions; concept drift changes the feature-target relationship

B. Both refer to the same thing

C. Data drift changes labels; concept drift changes features

D. Neither affects models

Answer 98

Correct: A — data drift changes inputs; concept drift changes the relationship.

Wrong:

B is false.

C reverses the definitions.

D is false.

Question 99

Which is a fairness concern when deploying ML?

A. Biased training data leading to discriminatory outcomes
B. Faster inference
C. Lower storage cost
D. Better dashboards

Answer 99

Correct: A — biased data can cause discriminatory outcomes.

Wrong:

B, C, D are performance or cost concerns, not fairness.

Question 100

Which practice best guards against leakage in a production pipeline?

A. Fit all transforms on the entire dataset

B. Encapsulate preprocessing and model in a Pipeline fit only on training data

C. Use the test set for feature selection

D. Scale before splitting

Answer 100

Correct: B — a Pipeline fit on training data prevents leakage.

Wrong:

A leaks test information.

C uses test data for selection.

D scales before splitting, leaking.